Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Predict Heart Disease Risk
2. Abbreviated Title	Predict Risk
3. Use Case Id	UC-03
4. Actors	Clinician (Primary), ML Engine (Supporting), MySQL Database (Supporting)
5. Description	The Clinician submits validated patient data to obtain a binary heart disease risk prediction (HIGH/LOW) with a probability score from the CNN-LSTM-Attention deep learning model. Includes preprocessing, model inference, and record storage.
5.1. Pre Conditions	1. All 13 patient features are entered and validated (UC-01). 2. A trained model checkpoint exists and is loaded. 3. Backend Flask server is running.
5.2. Task Sequence	1. Clinician clicks 'Run Diagnostic Scan' after entering patient data. 2. System transitions to the Processing Overlay (scanning animation). 3. System invokes UC-10: Preprocess and GAF Transform on the input data. 4. System invokes UC-11: Run CNN-LSTM Attention Model for inference. 5. System invokes UC-15: Manage Patient Records to store the result. 6. System transitions to the Results Dashboard. 7. System displays risk label (HIGH/LOW), probability score, and metrics.
5.3. Post Conditions	1. Risk prediction (probability and label) is displayed on the dashboard. 2. Prediction record is saved to the MySQL database. 3. XAI analysis is triggered automatically.
6. Modification History	Date: September 11, 2026
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